

Prithvish Baidya

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Blockchain infrastructure engineer. Co-authored [ERC-7518](#). Built the transaction backend powering [American Express's NFT program](#). Won [ETHIndia 2024](#) (2,500+ hackers, 36 hours). 4 years shipping production systems: transaction infrastructure, key management, and protocol tooling.

EXPERIENCE

thirdweb | Software Engineer

Jul 2024 – Nov 2025

Remote

- Built [engine-core](#): the transaction backend powering thirdweb's API. Enables one wallet to send thousands of transactions without nonce collisions, the hard problem that makes "mint 10,000 NFTs" actually work. Powers [Amex Passport](#), issuing digital travel stamps to millions of cardholders.
- Designed the core scheduling architecture: each wallet runs as a single unit of work in a distributed scheduler. One worker owns the entire lifecycle. No distributed locks, no coordination complexity, just correct ordering at scale.
- Built [twmq](#): Redis-backed durable job queue with exactly-once delivery. Handles crash recovery, retries, and job lifecycle tracking for engine-core.
- Designed **Vault**: non-custodial key management using AWS Nitro Enclaves. Keys live in a TEE with cryptographic attestation proving they never leave, even from thirdweb. Enterprise-grade custody without the custody.

ZoniqX | Lead Architect

Jul 2022 – Jun 2024

Remote

- Co-authored [ERC-7518](#): Ethereum standard for compliant real-world asset tokenization. Introduces partition-based semi-fungible tokens with dynamic transfer restrictions, the missing primitive for securities on-chain.
- Built the first compliant RWA tokenization layer on XRPL.
- Architected enterprise platforms: **Heimdall** (web3 operational layer) and **CompliTO** (compliant security token transfers).

Biconomy | Software Engineer

Nov 2021 – May 2022

Remote

- Rebuilt gasless transaction infrastructure processing millions of meta-transactions. Enterprise clients included JP Morgan.
- Built the first multichain **ERC-4337** smart contract wallet (hackathon POC turned production).

Libertas Network | Senior Fullstack Developer

Mar 2021 – Nov 2021

Remote

- Built decentralized censorship-resistant podcasting platform using IPFS, Node.js, and Ethereum.

PROJECTS

[zkRail](#) | [ETHIndia 2024 Winner](#) | [Demo Video](#)

Dec 2024

Smart contracts can't pay your landlord. zkRail fixes this: trustless fiat settlement using ZK proofs over banking APIs. Solver network executes real payments; game-theoretic bonds prevent fraud. Won 1st place overall (2,500+ participants) plus 4 track bounties (Polygon, Coinbase, CDP, Base). Built in 36 hours. [Technical writeup](#).

[t2z](#) | 1st Place, Electric Coin Company Track, Zyperpunk 2025

2025

Rust library for Zcash shielded transactions. Cross-platform via UniFFI: TypeScript (WASM), Go, Kotlin. Implements ZIP-374 PCZT for multi-party transaction construction. No proving key downloads; generates keys programmatically with Halo 2.

[Coredrain](#)

2025

Indexer for Hyperliquid's exotic architecture (HyperCore L1 to HyperEVM). Correlates L1 transfers with EVM transaction hashes. Discovered [undocumented dual-hash convention](#) for system transactions by reverse-engineering nanoreth source. Lazy indexing with block interpolation search.

PUBLICATIONS & OPEN SOURCE

[ERC-7518: Dynamic Compliant Interop Security Token](#), Co-author

Review, Ethereum EIPs

[engine-core](#): Production Rust transaction infrastructure for blockchain. Primary author.

2024–2025

[Bun, PR #25514](#): Added S3 Requester Pays support to Bun's native S3 client. Zig, TypeScript.

2025

RECOGNITION

[ETHIndia 2024](#): 1st place overall (2,500+ hackers) plus Polygon, Coinbase, CDP, Base bounties

Dec 2024

Zypherpunk 2025: 1st place, Electric Coin Company track

2025

KVPY-SA Fellowship: National science fellowship by Indian Institute of Science. Top 1% of 100,000+ applicants selected for research mentorship.

2019

NTSE Scholar: National Talent Search Examination. Nationwide exam selecting top 1,000 students from 1.5M+ for merit scholarship.

2019